

proposals for the third project

1 The third Project

- presentations in week fifteen

2 Topics

- ordinary differential equations; dynamical systems
- frequency domain methods; control systems
- partial differential equations

MCS 472 Lecture 35
Industrial Math & Computation
Jan Verschelde, 10 April 2026

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The Third Project

We are at the end of week twelve.

- Presentations happen in week fifteen.
- Work on the project in weeks thirteen and fourteen.
- Commit to a project topic soon.
- Preferably by 5pm on Monday 13 April.
- Send an email with the address of your partner in the CC.

proposals for topics

Many lectures ended with proposals.

- 1 ordinary differential equations; dynamical systems
- 2 frequency domain methods; control systems
- 3 partial differential equations; heat and traffic

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1. population model with age distribution (L-30)

Proposal for a Topic of a Project:

One deficiency of population models is that they do not take the age structure into account.

Read section 1.7 of *Mathematical Biology I. An Introduction* by James D. Murray, Springer-Verlag 2002.

Consider the following.

- Work through section 1.7 computationally and build a numerical model to reproduce the plots in that section.
- What is the effect of an aging population on its growth?

2. a discrete model for a population (L-30)

Proposal for a Topic of a Project:

Let a discrete model for a population N_k be given by

$$N_{k+1} = \frac{r N_k}{1 + b N_k^2}, \quad k = 0, 1, \dots$$

where r and b are positive parameters.

- 1 Using numerical values for r and b , run simulations of this model and interpret the role of r and b in the growth and/or decay of N_k .
- 2 Show that for any r , the population grows extinct if $b > 4$.
- 3 Determine the steady states and their eigenvalues.
Show that $r = 1$ is a bifurcation value.

Adapted from exercise 7 of Chapter 2 of *Mathematical Biology I. An Introduction* by James D. Murray, Springer-Verlag 2002.

3. expressway traffic flow (L-31)

Proposal for a Topic of a Project:

Take data on expressway traffic.

At various times of the day:

- What is the separation between cars?
- What is the separation versus velocity?
- How is flow related to density?

When is the road most dangerous?

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4. stability of following cars (L-33)

Proposal for a Topic of a Project:

Consider the model for two vehicle traffic

$$\frac{d}{dt}y(t + \tau) + \alpha y(t) = \alpha u(t).$$

- Obtain the stability condition for this problem.

In particular, what is the relation between τ and α for this problem to be stable?

- Show that even when stability is satisfied, a long platoon of vehicles is inherently unsafe because changes in speed of the lead car produce larger and larger deviations for cars farther down the platoon.

5. social feedback systems (L-34)

Supply and demand can be described via feedback.

Proposal for a Topic of a Project:

Read the paper by Otto Mayr on “Adam Smith and the Concept of the Feedback System: Economic Thought and Technology in 18-th Century Britain” in *Technology and Culture*, Vol. 12, No. 1, 1971.

Adam Smith suggested two processes:

- 1 Workers compare various employments and stream into that employment which offers the greatest rewards.
- 2 The rewards diminish as the number of competing workers rises.

Do the following:

- Use feedback to describe the labor market.
- Apply your description to the academic labor market.
- Find data comparing mathematics and computer science.

6. software for control systems (L-34)

Scilab and Xcos are open source software packages to design and analyze control systems.

Proposal for a Topic of a Project:

Visit `scilab.org`, download and install.

Do the following:

- Give an overview of the capabilities of the software.
- Reproduce one of the use cases listed on the web site.
- Write a report on your computational experiments.

7. design an automobile cruise control system (L-35)

Our text book has the design of an automobile cruise control system.

Proposal for a Topic of a Project:

Study the solution to this design problem in our text book.

Do the following:

- Describe the solution using your own words.
- Do the computational experiments to examine the stability.
- Write a report on your computational experiments.

8. root finding with the winding number (L-35)

The winding number leads to a numerical method to approximate all roots of a general nonlinear equation in a domain of the complex plane.

Proposal for a Topic of a Project:

Study the proof of the argument principle theorem in our text book and make sure you understand the definition of the winding number.

Do the following:

- Describe the algorithm to compute all complex roots in a domain of a general nonlinear function.
- Experiment with the implementation in Pari/GP.
- Write a report on your computational experiments.

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9. numerically solve the advection diffusion equation (L-36)

The advection diffusion equation is solved in chapter 2 of *Industrial Mathematics: A Course in Solving Real-World Problems* by Avner Friedman and Walter Littman, published by SIAM in 1994.

Proposal for a Topic of a Project:

Study chapter 2 of the Friedman-Littman book.

Review section 8.1 of *Numerical Analysis* of T. Sauer (MCS 471).

Do the following:

- Describe the solution using your own words.
- Do the computational experiments to examine the stability.
- Write a report on your computational experiments.

10. Fick's laws of diffusion (L-36)

The heat equation was derived by Adolf Fick in 1855.

Proposal for a Topic of a Project:

Study the wikipedia page on the derivation of Fick's second law and consult the references.

Do the following:

- Describe the derivation of the heat equation.
- Reproduce the illustration on the wikipedia page on Fick's laws of diffusion.
- Write a report on your computational experiments.

11. population density of age segment (L-36)

We are interested to model the population of an age segment.

Proposal for a Topic of a Project:

Let $u = u(x, t)$ be the population density defined as

$$P(b, t) - P(a, t) = \int_a^b u(x, t) dx$$

the population at time t between the ages a and b . Let $q(x)$ be the probability of death less net immigration at age x per year.

- Show that $u_t + u_x + qu = 0$.
- Solve analytically (see page 227 of the text book).
- Simulate the model numerically and interpret the results.

12. the Fourier ring problem (L-37)

A long rod is bent and its ends welded to form a ring. Given an initial temperature profile around the ring, how will this temperature profile evolve over time?

Proposal for a Topic of a Project:

Study the solution of Problem 3 in our text book.

Do the following:

- Describe the solution using your own words.
- Do the Fourier series computations.
- Write a report on your computational experiments.

13. the heat diffusion problem in polar coordinates (L-37)

Consider the preparation for a camping trip.
To not freeze to death during the night in a tent,
a heater for tent camping is needed.

Proposal for a Topic of a Project:

Our tent has a circular base, with the heater at its center.
Develop a mathematical model to prepare for a safe camping trip.

Do the following:

- Gather data on tent heaters.
- Describe the heat diffusion equation in polar coordinates.
- Interpret the computational results with the data.

14. least squares fitting (L-38)

The models are experimentally verified via least squares.

Proposal for a Topic of a Project:

Apply least squares fitting on observed traffic data.

Do the following:

- Collect data from an real highway.
- Fit the data for $k(v)$ and $q(v)$.
- Examine how realistic the models are.

In particular, can your model

- constructed with the data from one day,
- predict the traffic of the next day?

15. geothermal heat pumps (L-39)

A geothermal heat pump compensates for the seasonal temperature fluctuations, to heat a home during the winter with the warmth of the earth stored during summer.

Proposal for a Topic of a Project:

- Use the Helmholtz phasor method to explain how deep the system must be buried.
- Do a cost benefit analysis of installing a geothermal heat pump, comparing to heating a home with a furnace.

16. thickness of walls for isolation (L-39)

Native Americans of the southwest built homes with adobe walls of a critical thickness. The walls delayed the flux of the sun until it was needed to heat the interior during the cold nights.

Proposal for a Topic of a Project:

- Model the thickness of the adobe walls.
- Use actual material properties to select the optimal thickness for a 24-hour repetitive cycle.
- Compare the results of your model against actual constructions.

17. study hibernating worms (L-39)

Birds migrate to survive the winter.

Worms have no wings but they do survive each winter.

Proposal for a Topic of a Project:

How deep do hibernating worms go to survive the winter?

- Gather biological data on worms.
- How deep does one have to dig to find worms?
- Use the Helmholtz method to answer the question.

18. Can you hear the shape of a drum? (L-40)

This question is the title of the paper by Mark Kac, published in April 1966 by *The American Mathematical Monthly*, pages 1-23.

Proposal for a Topic of a Project:

This problem is still *open*, according to the paper “Can’t One Really Hear the Shape of a Drum?” by S. Zuluaga and F. Fonseca, in *Acoustical Physics*, Vol. 57, No. 4, pages 465–472, 2011.

- Study the papers and illustrate the importance of this problem for industrial applications.
- Summarize the findings of the two papers.